

Conversion Log

Production Conversion Log

January 2025 · Girardot, Colombia

Submission reference	RTF0-310125
Reporting period	January 2025 (2025-01)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,780 kg/L
Density — PLUS-grade	0,856 kg/L
Production days observed	21 of 31 (10 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-20T00:00:00Z)
Generated	2026-05-20T00:00:00Z

Signed for and on behalf of OisteBio GmbH

[CEO_NAME]

Chief Executive Officer

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **January 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 043 312,000 kg
Total EU-grade production	760 828,000 kg / 975 420,513 L
Total PLUS-grade production	1 308 624,160 kg / 1 528 766,542 L
Liquid-product yield (EU + PLUS) over input	68,00 %
Grand total volume (EU + PLUS)	2 504 187,055 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **426 063,680 kg**, steel scrap **286 071,328 kg**, water condensate **51 736,304 kg**, syngas **144 557,318 kg**, recorded losses **65 431,203 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — January 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,780 kg/L, PLUS 0,856 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-01-04	121 242,000	30 310,500	52 134,060	0,780	0,856	38 859,615	60 904,276	68,00 %	—
2025-01-06	137 180,000	34 295,000	58 987,400	0,780	0,856	43 967,949	68 910,514	68,00 %	—
2025-01-07	150 632,000	37 658,000	64 771,760	0,780	0,856	48 279,487	75 667,944	68,00 %	—
2025-01-08	140 093,000	35 023,250	60 239,990	0,780	0,856	44 901,603	70 373,820	68,00 %	—
Total	3 043 312,000	760 828,000	1 308 624,160	—	—	975 420,513	1 528 766,542	68,00 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-01-09	145 235,000	36 308,750	62 451,050	0,780	0,856	46 549,679	72 956,834	68,00 %	—
2025-01-10	146 915,000	36 728,750	63 173,450	0,780	0,856	47 088,141	73 800,759	68,00 %	—
2025-01-13	158 292,000	39 573,000	68 065,560	0,780	0,856	50 734,615	79 515,841	68,00 %	—
2025-01-14	147 889,000	36 972,250	63 592,270	0,780	0,856	47 400,321	74 290,035	68,00 %	—
2025-01-15	141 214,000	35 303,500	60 722,020	0,780	0,856	45 260,897	70 936,939	68,00 %	—
2025-01-16	148 245,000	37 061,250	63 745,350	0,780	0,856	47 514,423	74 468,867	68,00 %	—
2025-01-17	136 842,000	34 210,500	58 842,060	0,780	0,856	43 859,615	68 740,724	68,00 %	—
2025-01-20	153 240,000	38 310,000	65 893,200	0,780	0,856	49 115,385	76 978,037	68,00 %	—
2025-01-21	152 708,000	38 177,000	65 664,440	0,780	0,856	48 944,872	76 710,794	68,00 %	—
2025-01-22	141 993,000	35 498,250	61 056,990	0,780	0,856	45 510,577	71 328,259	68,00 %	—
2025-01-23	151 976,000	37 994,000	65 349,680	0,780	0,856	48 710,256	76 343,084	68,00 %	—
2025-01-24	140 403,000	35 100,750	60 373,290	0,780	0,856	45 000,962	70 529,544	68,00 %	—
2025-01-27	146 115,000	36 528,750	62 829,450	0,780	0,856	46 831,731	73 398,890	68,00 %	—
2025-01-28	160 323,000	40 080,750	68 938,890	0,780	0,856	51 385,577	80 536,086	68,00 %	—
2025-01-29	142 639,000	35 659,750	61 334,770	0,780	0,856	45 717,628	71 652,769	68,00 %	—
2025-01-30	153 252,000	38 313,000	65 898,360	0,780	0,856	49 119,231	76 984,065	68,00 %	—
2025-01-31	126 884,000	31 721,000	54 560,120	0,780	0,856	40 667,949	63 738,458	68,00 %	—
Total	3 043 312,000	760 828,000	1 308 624,160	—	—	975 420,513	1 528 766,542	68,00 %	

Non-production days (10): 2025-01-01, 2025-01-02, 2025-01-03, 2025-01-05, 2025-01-11, 2025-01-12, 2025-01-18, 2025-01-19, 2025-01-25, 2025-01-26. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia for January 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at constant product density measured at 15 °C. The two nominal densities used in this report are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,780 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,856 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family and are loaded into the platform `product_densities` seed table by Alembic migration `0005_seed_product_densities.py`. They are immutable for the duration of a reporting period and are re-validated against laboratory analysis at every product release.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.78) STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.856) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310125 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix `000000000000`). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

— End of Production Conversion Log —